Tower of Hanoi

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Awaking from your 5 hour sleep, you emerge from your bed dreading today—and then you realize it's Monday, the day you're supposed to present for your history class. Not only are you unprepared, you haven't even started your slideshow or opened up the rubric. Because you're a math nerd who takes AP French, you immediately come up with a presentation idea: **The Tower of Hanoi**.

The premise of the puzzle is to find the minimum possible number of moves to move n **discs**, which are strictly decreasing in size from top to bottom, from one of k **pegs** to another with the following restrictions:

- Discs may only be transferred between pegs one move at a time.
- No larger disc may be placed on top of a smaller disc.
- Only the top disc of a peg may be moved to another peg.
- Define a move to be a transfer of a disc from one peg to another.

The most common variation of this puzzle involves 3 pegs and 3 discs, but it is nearly just as likely to see 3 pegs and any other positive integer n.

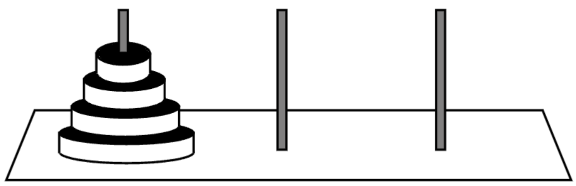

Figure 1: Unsolved tower with 3 pegs and 4 discs. (Wolfram MathWorld)

Questions:

1. What is the minimum number of moves requires to transfer 3 discs to another peg given that there are 3 pegs?
2. Let $H(n, k)$ represent the minimum number of moves to transfer n discs given k pegs. What are $H(4, 3)$? $H(5, 3)$? $H(10, 3)$?
3. Express $H(n, 3)$ in terms of n.
4. Consider $H(x, x)$.

(a) Are there any restrictions for the outputs of $H(x, x)$? Do these encompass all the restrictions of $H(n, k)$?
(b) What is $H(4, 4)$? What about $H(5, 5)$? $H(10, 10)$?
(c) Express $H(x, x)$ in terms of x.

5. Is it possible to solve the standard Tower of Hanoi in an even number of moves? The moves do not necessarily have to be optimal.

6. Define $g(n) = \min(H(n, k))$ as k varies.

   (a) Find $g(3)$, $g(5)$, and $g(15)$.
   (b) What is the minimum value of k for which $g(n) = H(n, k)$?
   (c) Express $g(n)$ in terms of n.

7. Consider a variation of this puzzle with the following additions:

   - Every disc now has an orientation; its top surface is black, and its bottom surface is white.
   - The original position of every disc has its black surface facing upwards.
   - Every move requires a reorientation of the disc onto its opposite surface.
   - The base of a peg is colored and counts as a surface.
   - At any point in the game, no two surfaces with the same color may be touching.

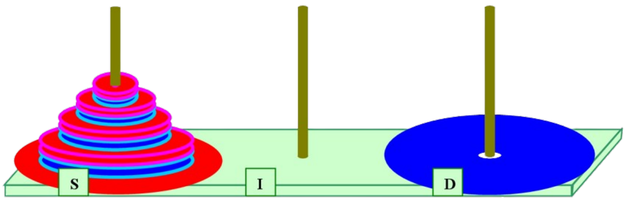

Figure 2: Magnetic tower, with a red side facing upwards, the base of the pegs are also colored here.

Define $S(n)$ to be the minimum number of moves required to move n discs with 3 pegs and consider the coloring of the pegs to be white-white-black or black-white-black.

   (a) Find $S(2)$, $S(3)$, and $S(5)$.
   Consider $S(3)$.
   (b) Express $S(n)$ in terms of n.

References

Levy, Uri. "The Magnetic Tower of Hanoi." *ArXiv* abs/1003.0225, 2010.
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Weisstein, Eric W. "Tower of Hanoi." From MathWorld–A Wolfram Web Resource, Feb 2024.
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The solution is on page 23.